# Proyecto Integrador

**Samantha Chamorro**

**Jeniffer Condor**

**Luis Quisnancela**

**Redes**

**Universidad Internacional Del Ecuador**

**Facultad de Ciencias Técnicas**

**Ingeniería en Sistemas de la Información**

**Bryan Steven Vinueza Bustamante**

Repositorio: https://github.com/sammy1208-g/Proyecto_Integrador_3_nivel

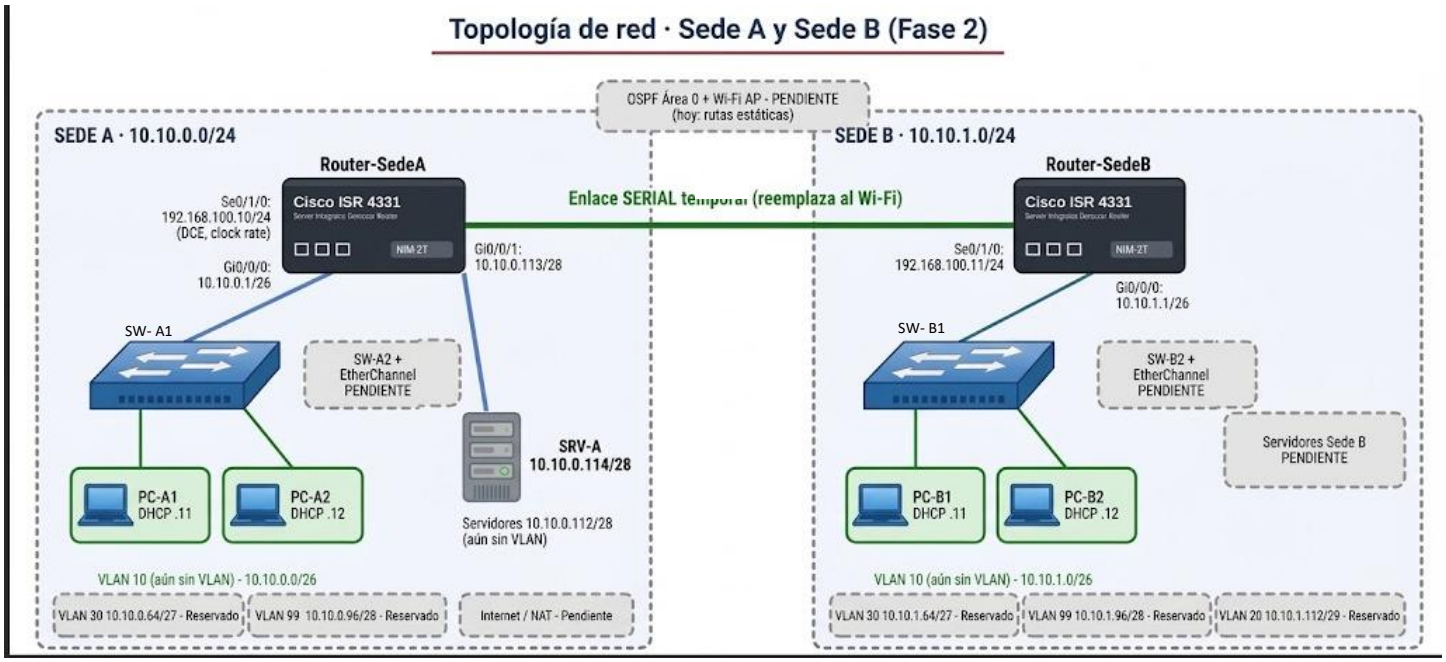
**1 de Noviembre del 2026**

# Phase 2: Packet Tracer Network Prototype

**Team Contributions**: This report was prepared by Group 6 during a joint working session in which the three members built and tested the Packet Tracer prototype together. Samantha Chamorro led the construction of the topology and the documentation. Luis Quisnancela and Jeniffer Condor took part in the whole process: they helped build and configure the network, reviewed each step against what was seen in class, and took part in the technical decisions, such as the choice of the ISR 4331 router model used in class and the decision to use a serial link between sites after testing other options.

## 1. Updated Architecture

**Diagram**




*Color = implemented · Dashed gray = planned (pending)*

***Figure 1**. Updated diagram*

The Phase 1 diagram was updated to match the Packet Tracer prototype. Implemented elements are shown in color and planned elements in gray with a dashed line. Three changes were made. First, the Wi-Fi link between sites was replaced by a temporary serial link, because Wi-Fi is not available in the simulator. Second, each site uses one switch and no VLANs yet, because VLANs are covered in Week 10. Third, the routers are Cisco ISR 4331 devices, the model used in class, while OpenWrt on Raspberry Pi remains the target platform.

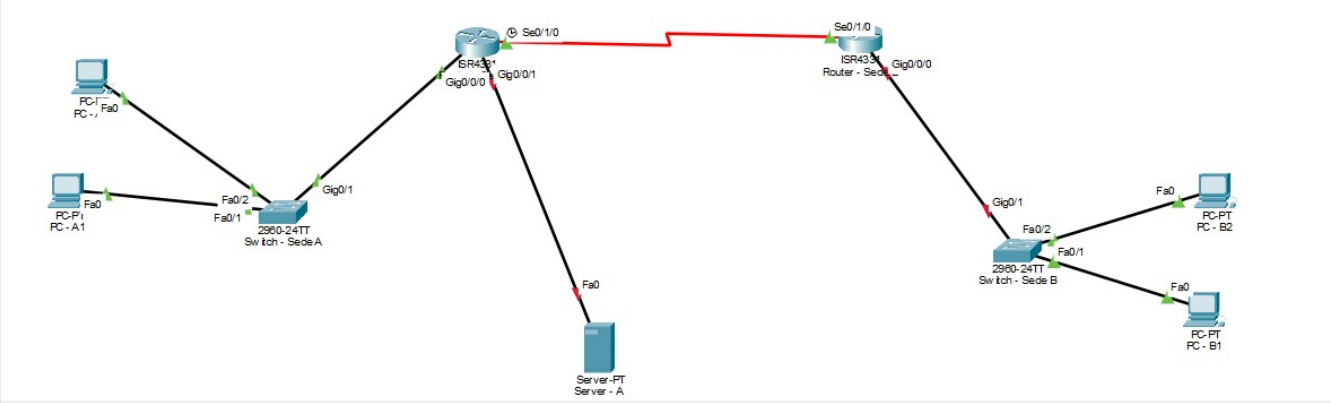



***Figure 2.*** *Topology of Packet Tracer*

## 2. VLSM Addressing

The addressing follows the Phase 1 plan from 10.10.0.0/16: block 10.10.0.0/24 for Site A and 10.10.1.0/24 for Site B. Subnets for the planned VLANs (guests, management and Site B servers) are reserved but not implemented. The inter-site link uses 192.168.100.0/24, as in Phase 1.

| Site | Purpose | Hosts | Network/Prefix | Mask | Usable Range | Broadcast | Gateway | Allocation | Status |
|---|---|---|---|---|---|---|---|---|---|
| A | Users (VLAN 10) | 31 | 10.10.0.0/26 | 255.255.255.192 | .1 – .62 | .63 | 10.10.0.1 | Fixed gateway; clients by DHCP | **Implemented (no VLAN yet)** |
| A | Guests (VLAN 30) | 21 | 10.10.0.64/27 | 255.255.255.224 | .65 – .94 | .95 | 10.10.0.65 | DHCP | Reserved |
| A | Administration (VLAN 99) | 9 | 10.10.0.96/28 | 255.255.255.240 | .97 – .110 | .111 | 10.10.0.97 | Fixed | Reservedo |
| A | Servers (VLAN 20) | 7 | 10.10.0.112/28 | 255.255.255.240 | .113 – .126 | .127 | 10.10.0.113 | Fixed (SRV-A: 10.10.0.114) | **Implemented (no VLAN yet)** |
| B | Users (VLAN 10) | 31 | 10.10.1.0/26 | 255.255.255.192 | .1 – .62 | .63 | 10.10.1.1 | Fixed gateway; clients by DHCP (.11 – .62) | **Implemented (no VLAN yet)** |
| B | Guests (VLAN 30) | 21 | 10.10.1.64/27 | 255.255.255.224 | .65 – .94 | .95 | 10.10.1.65 | DHCP | Reserved |
| B | Administration (VLAN 99) | 9 | 10.10.1.96/28 | 255.255.255.240 | .97 – .110 | .111 | 10.10.1.97 | Fixed | Reservado |
| B | Servers (VLAN 20) | 3 | 10.10.1.112/29 | 255.255.255.248 | .113 – .118 | .119 | 10.10.1.113 | Fixed | Reserved |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Link | Inter-side (serial) | 2 | 192.168.100.0 /24 | 255.255.255.0 | .10 and .11 | — | — | Fixed (A: .10, B: .11) | **Implemented (temporary)** |

## 3. Basic Device Configuration

All devices were named (Router-SedeA, Router-SedeB, SW-A1, SW-B1), their interfaces were configured and activated, and the configurations were saved. Each switch connects its two PCs and its router through access ports, with all ports in VLAN 1 for now. The MAC address tables show that each switch learned the PCs on Fa0/1 and Fa0/2 and the router on Gig0/1. The server does not appear in any table because it connects directly to the router.

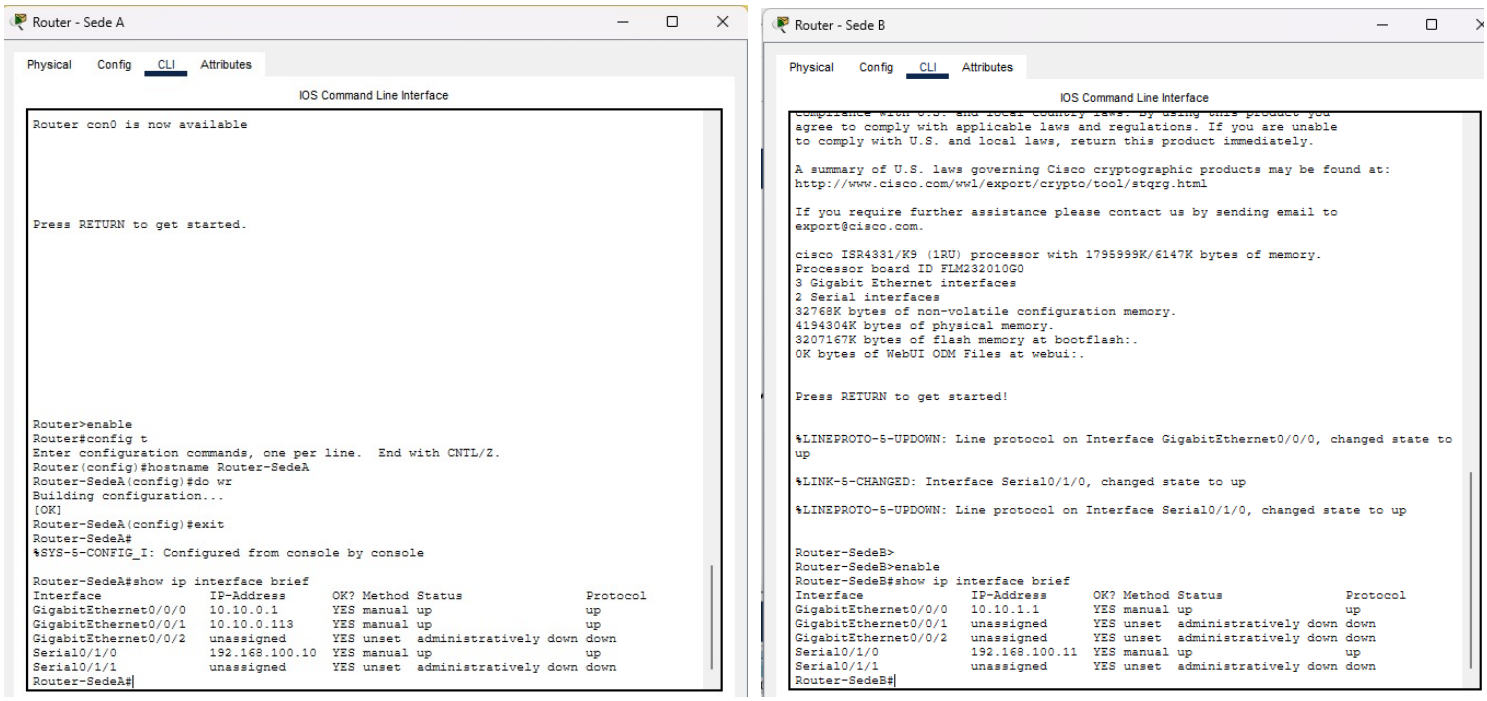



***Figure 3.*** *Command execution: show ip interface brief on Router-SedeA and Router-SedeB. The show ip interface brief command lists each router interface in one line with its IP address and its physical (Status) and link (Protocol) states, so you can quickly verify the configuration and confirm the links are active.*

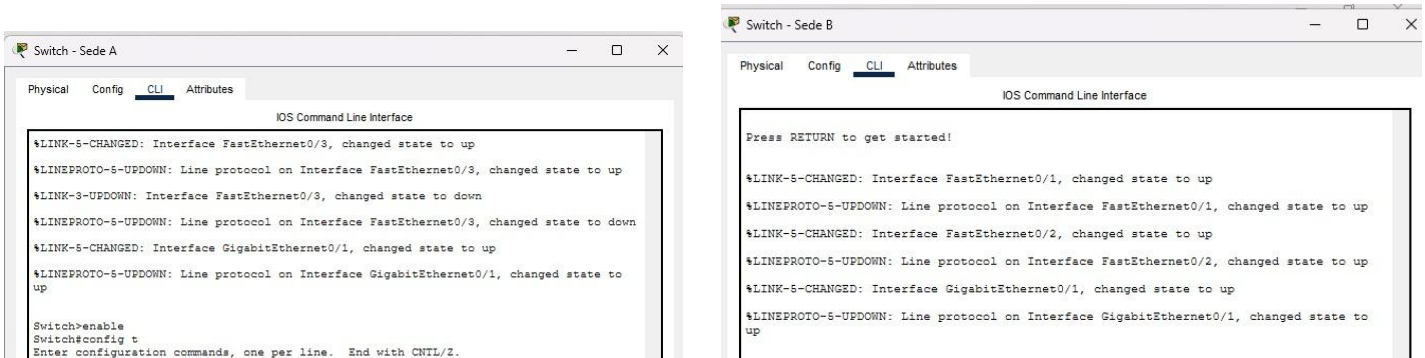


***Figure 4.*** *Command execution:* ***show mac address-table*** *on Switch A and B. The show mac address-table command lists the MAC addresses the switch has learned dynamically, along with the VLAN and port where each one was learned.*

## 4. DHCP

Each router runs a DHCP pool for its user network. Addresses .1 to .10 are excluded so they can be kept for the gateway and for fixed addresses. The pool assigns addresses from .11. The default gateway is the router interface and the DNS server is 8.8.8.8. All four PCs received valid addresses.

Excluding .1 to .10 keeps those addresses free for the gateway and for devices with fixed addresses, so the pool cannot give them to a client by mistake. Each pool starts at .11, which is why the first two clients in each site received .11 and .12. The DNS server 8.8.8.8 is set in the pools, but it was not tested because the prototype has no Internet access. All four PCs requested an address and received one from the router of their own site.

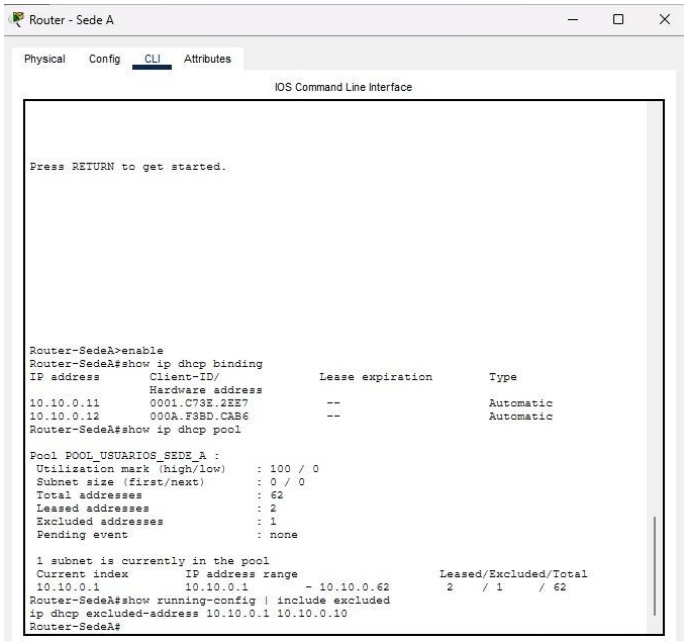
**POOLS table**

| Router | Pool | Network | Excluded Address | Gateway | DNS |
|---|---|---|---|---|---|
| **Router-SedeA** | POOL_USUARIOS_SEDE_A | 10.10.0.0/26 | 10.10.0.1 – 10.10.0.10 | 10.10.0.1 | 8.8.8.8 |
| **Router-SedeB** | POOL_USUARIOS_SEDE_B | 10.10.1.0/26 | 10.10.1.1 – 10.10.1.10 | 10.10.1.1 | 8.8.8.8 |

**Table of Received Addresses**

| Client | MAC | IP address received | Mask | Gateway |
|---|---|---|---|---|
| **PC-A1** | 0001.C73E.2EE7 | 10.10.0.11 | 255.255.255.192 | 10.10.0.1 |
| **PC-A2** | 000A.F3BD.CAB6 | 10.10.0.12 | 255.255.255.192 | 10.10.0.1 |
| **PC-B1** | 0030.F21B.5416 | 10.10.1.11 | 255.255.255.192 | 10.10.1.1 |
| **PC-B2** | 0002.4AB3.5526 | 10.10.1.12 | 255.255.255.192 | 10.10.1.1 |

DHCP was tested only on the two implemented user networks. DHCP for guests (VLAN 30) is planned and pending, because VLANs are not implemented yet.



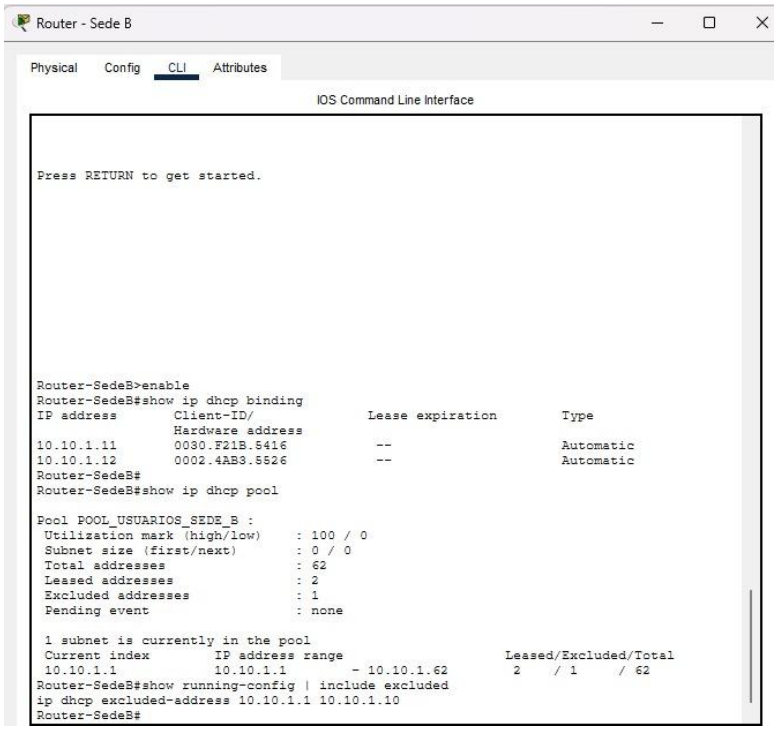



***Figure 5.*** *DHCP bindings, pool status and excluded range on Router-SedeA and Router-SedeB. IP configuration received by PC-A1, PC-A2, PC-B1 and PC-B2.*

## 5. Initial Intersite Routing

The two sites are connected for now with static routes over the serial link (192.168.100.0/24). Router-SedeA uses 192.168.100.10 and Router-SedeB uses 192.168.100.11. This is enough to test communication between sites in Phase 2, before moving to OSPFv2 in Phase 3.

| Router | Destination network | Next hop | Reason |
|---|---|---|---|
| **Router-SedeA** | 10.10.1.0/26 (Site B users) | 192.168.100.11 | Reach Site B |
| **Router-SedeB** | 10.10.0.0/26 (Site A users) | 192.168.100.10 | Reach Site A users |
| **Router-SedeB** | 10.10.0.112/28 (Site A servers) | 192.168.100.10 | Reach SRV-A |

Routing tables. Executing show ip route displays directly connected subnets (coded C and L) alongside static routes (coded S). Router-SedeA lists S 10.10.1.0/26 [1/0] via 192.168.100.11, while Router-SedeB lists the two static routes pointing toward Site A subnets.

Packet path. Running tracert from PC-B1 to SRV-A (10.10.0.114) traces three sequential hops: 10.10.1.1 (Router-SedeB), 192.168.100.10 (Router-SedeA), and 10.10.0.114 (SRV-A). This trace verifies end-to-end traversal from Site B to the server in Site A across the point-to-point link.

Connectivity tests. PC-A1 reached PC-B1 (10.10.1.11) with one lost packet in the first attempt, which is normal while the routers learn the MAC addresses, and 4 of 4 replies when repeated. PC-B1 reached SRV-A (10.10.0.114) with 4 of 4 replies.

Why static routes are temporary. Static routes must be written by hand on each router and do not change if the network changes. In Phase 3 they will be replaced by OSPFv2 in Area 0, so each router learns the other site’s networks automatically.

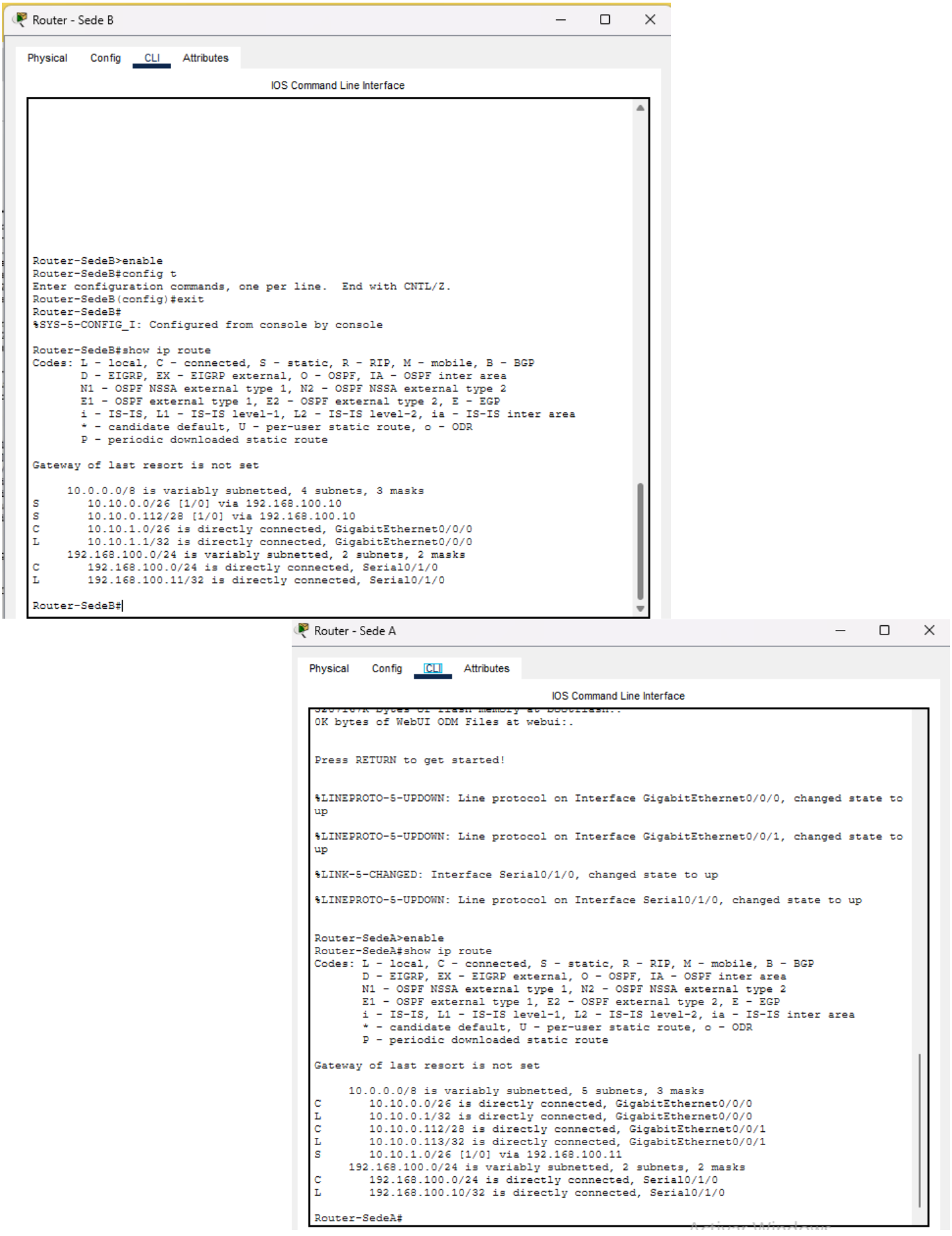



***Figure 6.*** *Routing table of Router-SedeA and Router-SedeB (show ip route).*

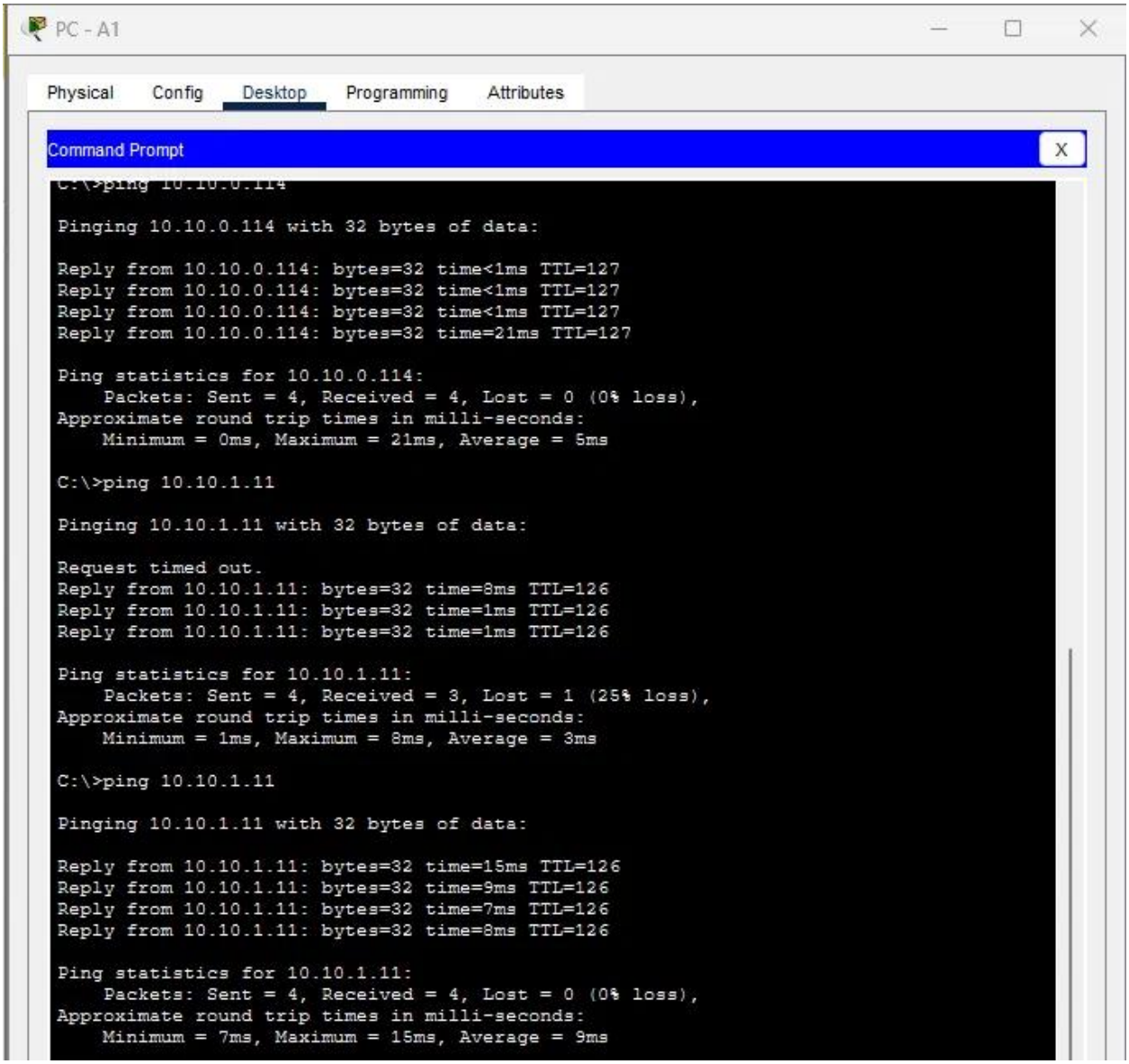



***Figure 7.*** *Ping from PC-A1 to SRV-A and to PC-B1.*

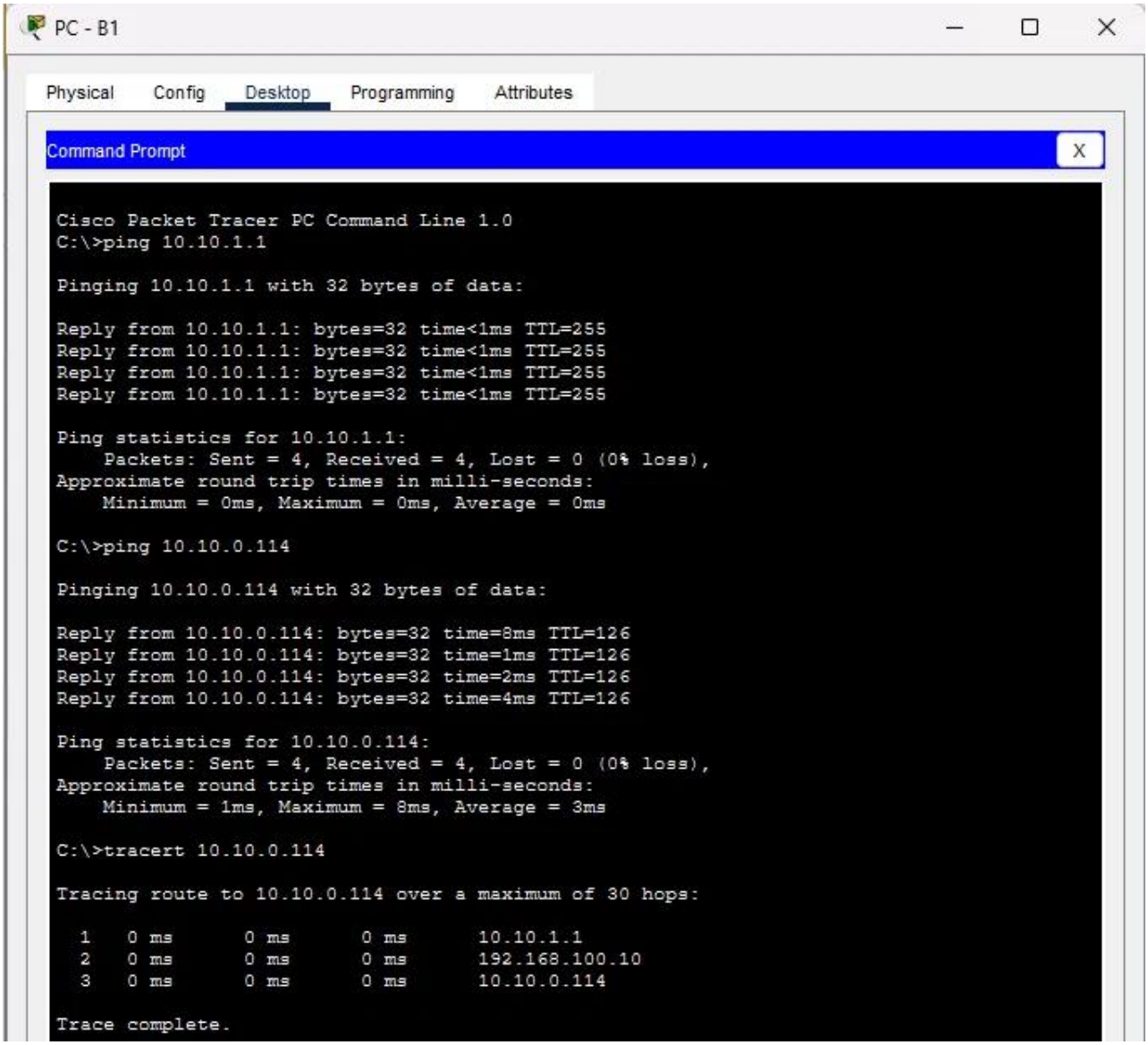



***Figure 8.*** *Ping and tracert from PC-B1 to SRV-A.*

## 6. Administration and Troubleshooting

In this phase, device management was conducted exclusively via the Packet Tracer local console. No passwords, tokens or private keys are included in this report or in the project repository. Secure remote management (SSH, a password for privileged mode, and access only from the management VLAN 99) is planned for the next phase.

| | |
|---|---|
| **Problem** | The link between sites was first planned on an Ethernet port of an added module (Gi0/1/0). |
| **Symptom** | Assigning an IP address to the port returned `% Invalid input detected`, and `no switchport` returned `% Incomplete command`. (Figure 9) |
| **Diagnosis** | `show ip interface brief` listed the module ports (Gi0/1/0 to Gi0/1/3). Because the router rejected an IP address on the port, we concluded it behaved as a switch port and not as a routed port. (Figure 10) |

| | |
|---|---|
| **Correction** | The module was replaced by a serial module (NIM-2T). The link was configured on Se0/1/0 in both routers, with `clock rate` on the DCE side (Router-SedeA). (Figure 11). |
| **Retest** | Both Se0/1/0 interfaces showed up/up, the static routes worked, and pings between sites succeeded (T-05, T-06, T-07). |

A workaround using a VLAN interface on the module was considered, but it was not used because that topic has not been covered in class yet.

A second issue was found during verification: the interface Gi0/0/1 (servers network) was administratively down because the no shutdown command had been missed, and the router names had not been applied. Both were corrected and verified with show ip interface brief.

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/1/0
Router(config-if)#no switchport
% Incomplete command.
Router(config-if)#int g0/1/0
Router(config-if)#ip add 192.168.100.10 255.255.255.0
                   ^
% Invalid input detected at '^' marker.

Router(config-if)#
Router#
```

*Figure 9. Errors returned when assigning an IP address to port Gi0/1/0.*

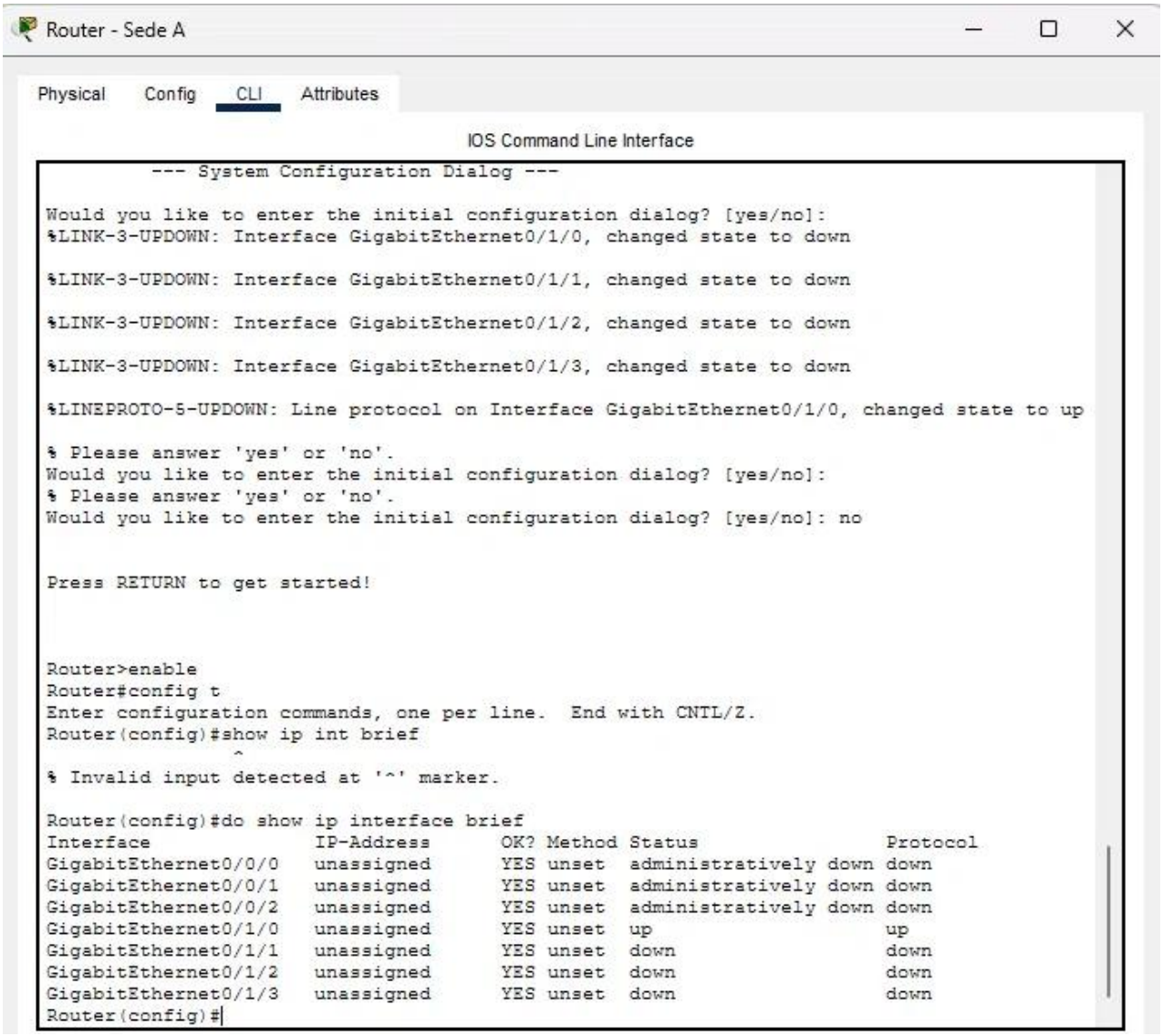



***Figure 10. show ip interface brief*** *showing the module ports Gi0/1/0 to Gi0/1/3 without an IP address.*

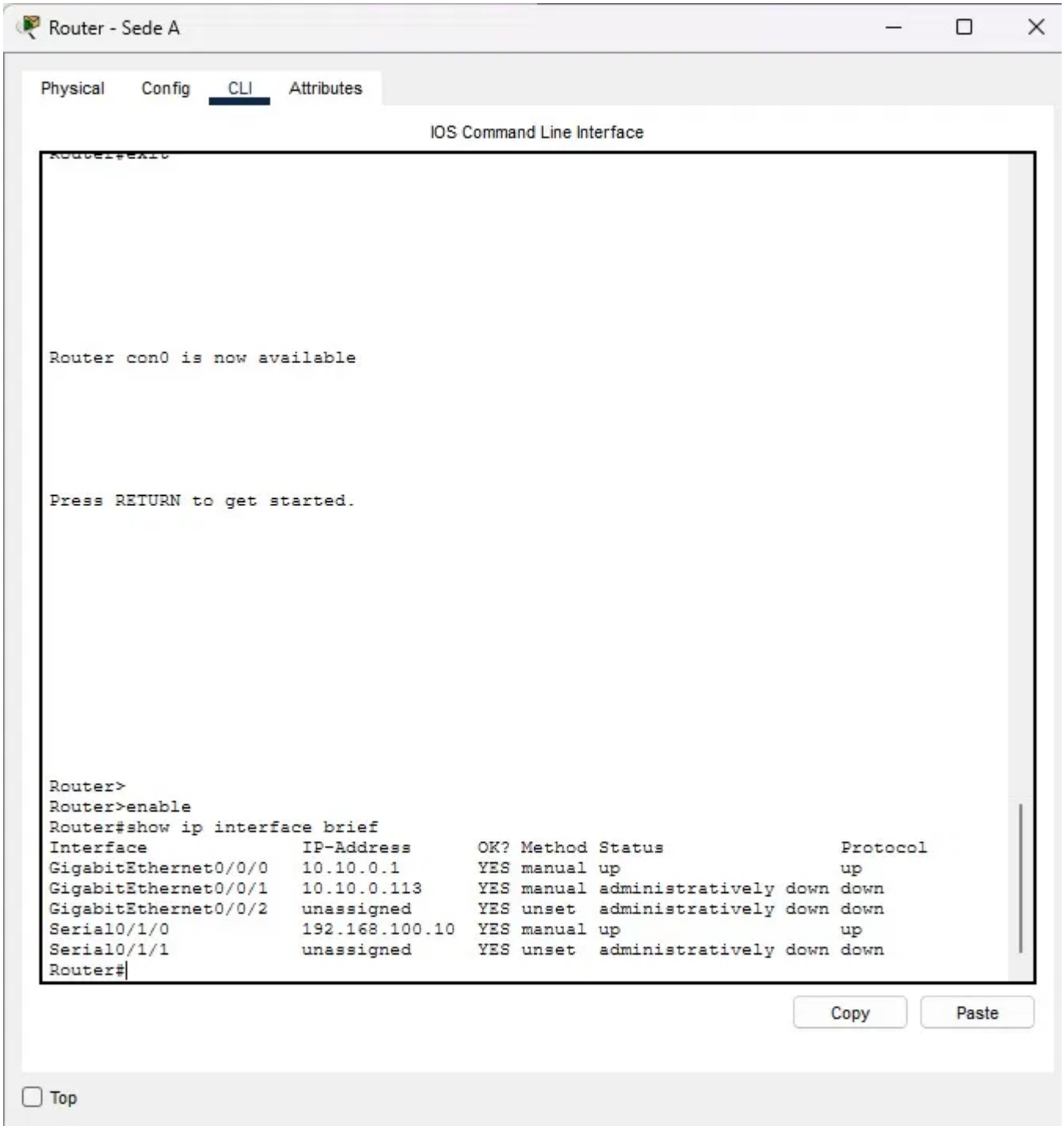
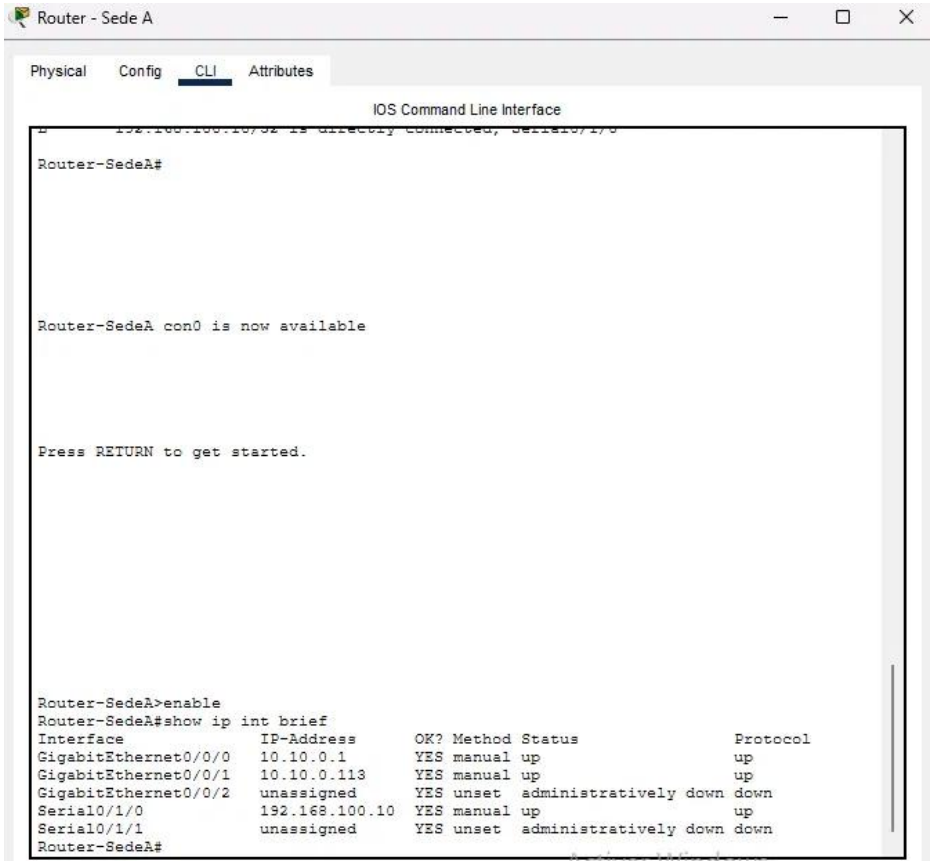



***Figure 11. show ip interface brief*** *showing Serial0/1/0 up/up and Gi0/0/1 administratively down.*


Router - Sede A
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router-SedeA#
Router-SedeA con0 is now available
Press RETURN to get started.
Router-SedeA>enable
Router-SedeA#show ip int brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 10.10.0.1 YES manual up up
GigabitEthernet0/0/1 10.10.0.113 YES manual up up
GigabitEthernet0/0/2 unassigned YES unset administratively down down
Serial0/1/0 192.168.100.10 YES manual up up
Serial0/1/1 unassigned YES unset administratively down down
Router-SedeA#

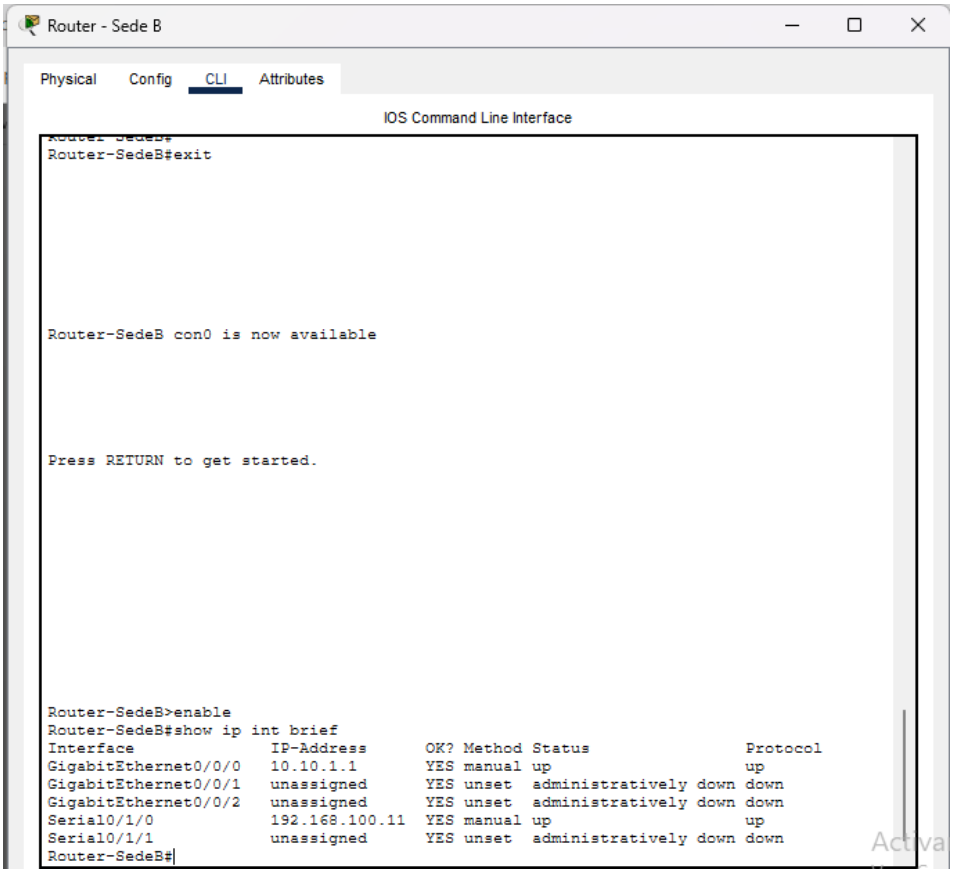
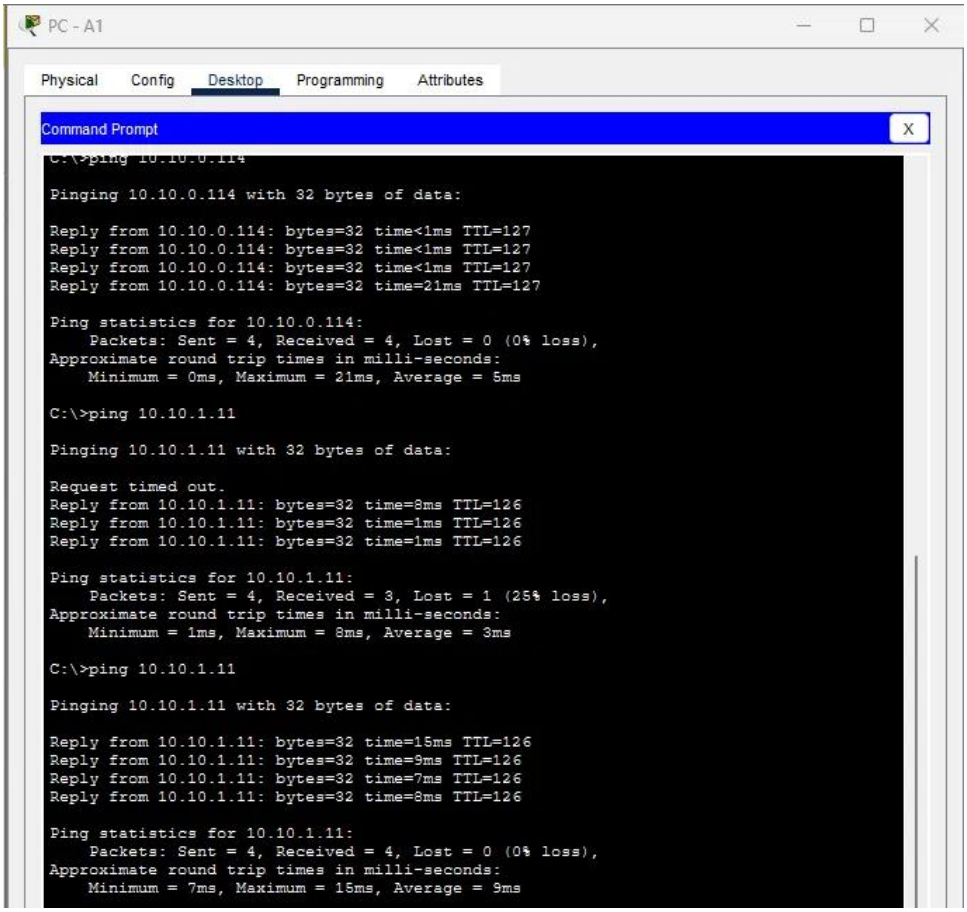

Router - Sede B
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router-SedeB#exit
Router-SedeB con0 is now available
Press RETURN to get started.
Router-SedeB>enable
Router-SedeB#show ip int brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 10.10.1.1 YES manual up up
GigabitEthernet0/0/1 unassigned YES unset administratively down down
GigabitEthernet0/0/2 unassigned YES unset administratively down down
Serial0/1/0 192.168.100.11 YES manual up up
Serial0/1/1 unassigned YES unset administratively down down
Router-SedeB#

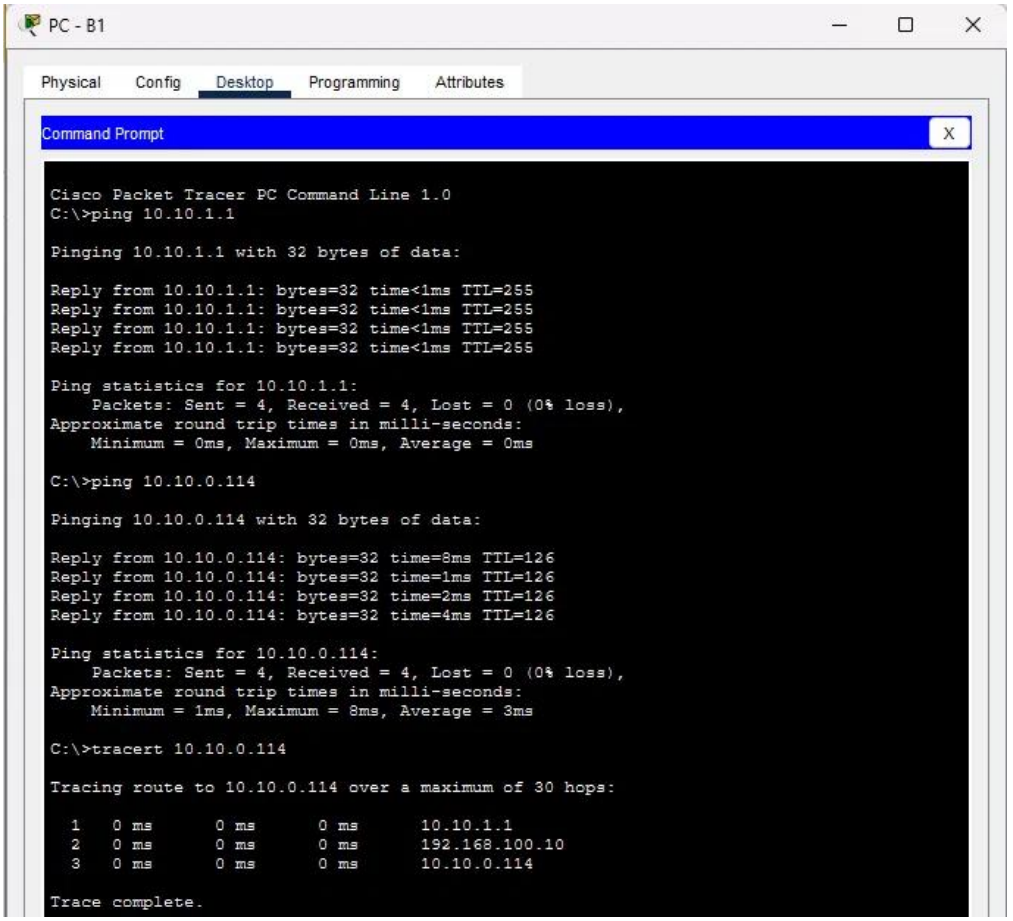
*Figure 12. Interface status of Router-SedeA and Router-SedeB after correction.*

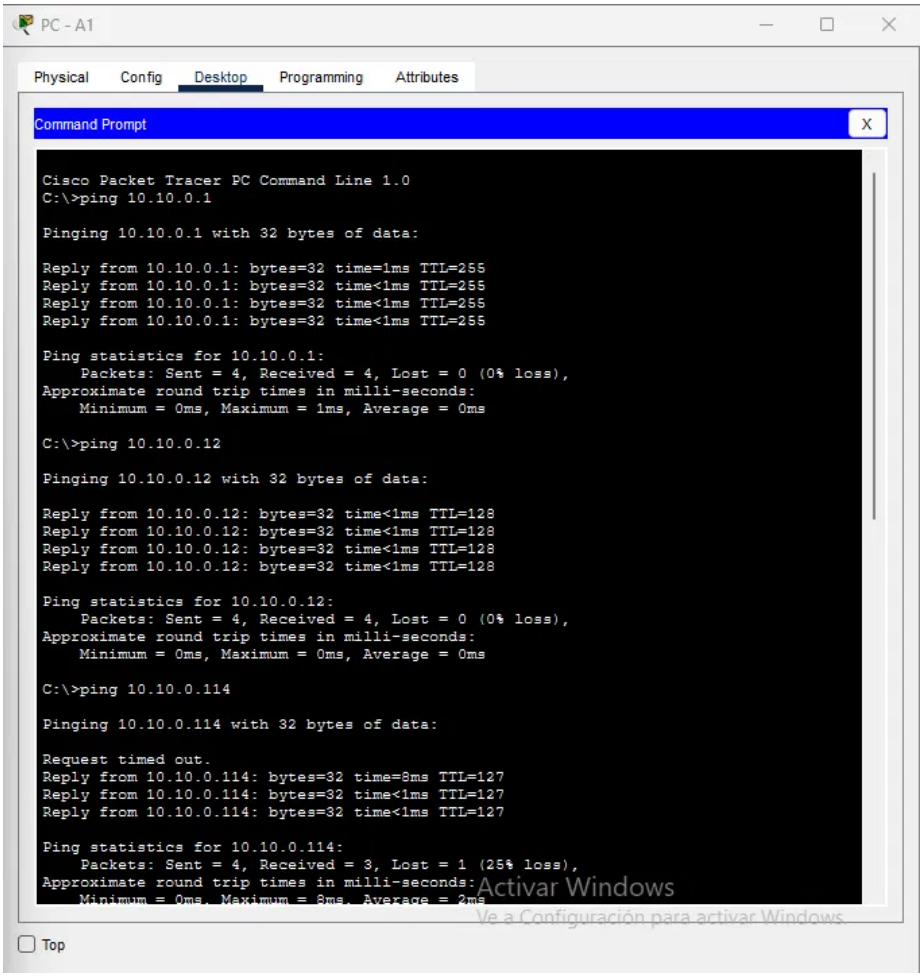
## 7. Preliminary Tests

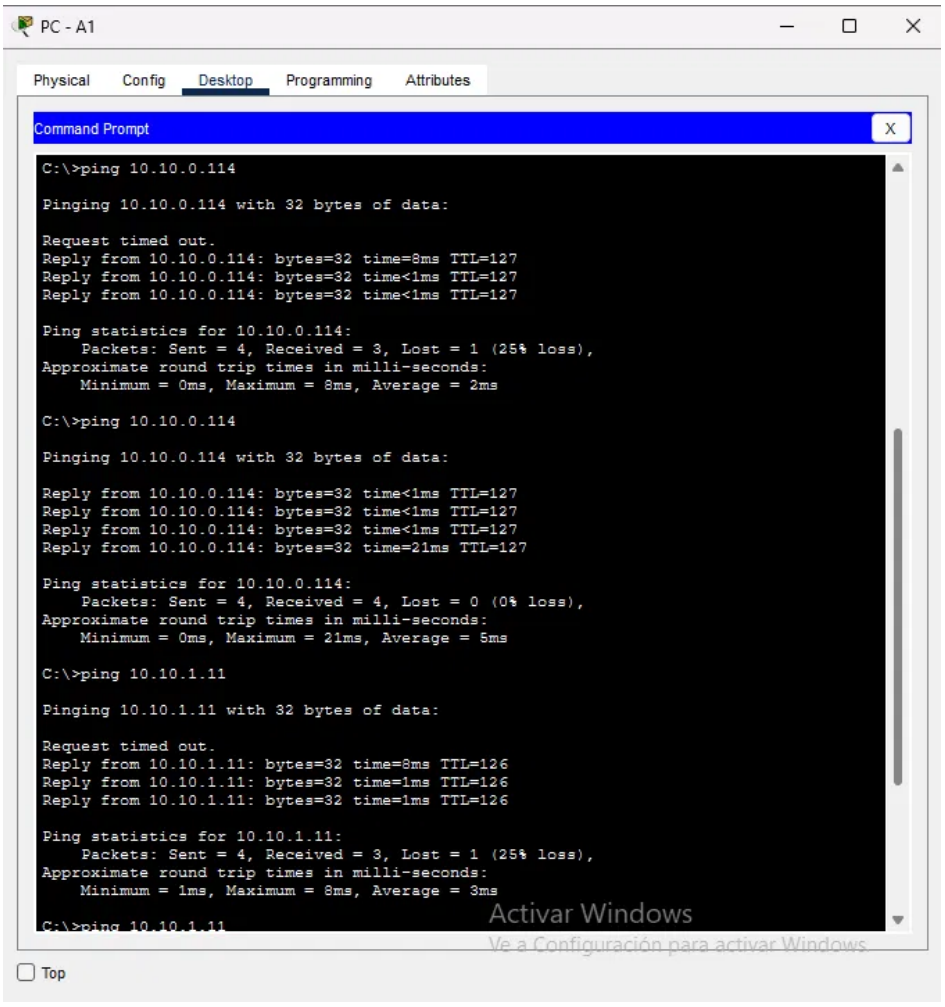
The tests below were run in the Packet Tracer prototype. Tests marked Pending depend on content planned for Week 10 and were not performed, so they are not counted as passed.

| Test ID | Source | Destination | Procedure | Expected result | Observed result | Evidence | Status |
|---|---|---|---|---|---|---|---|
| T-01 | PC-A1 | Gateway 10.10.0.1 | `ping 10.10.0.1` | 4 of 4 replies | 4 of 4 replies | Figure 7 | Passed |
| T-02 | PC-A1 | PC-A2 (10.10.0.12) | `ping 10.10.0.12` | 4 of 4 replies | 4 of 4 replies | Figure 7 | Passed |
| T-03 | PC-A1 | SRV-A (10.10.0.114) | `ping 10.10.0.114` | 4 of 4 replies | 3 of 4 in the first attempt (first packet lost while the MAC address was learned); 4 of 4 when repeated | Figure 7 | Passed |
| T-04 | PC-B1 | Gateway 10.10.1.1 | `ping 10.10.1.1` | 4 of 4 replies | 4 of 4 replies | Figure 8 | Passed |
| T-05 | PC-A1 | PC-B1 (10.10.1.11) | `ping 10.10.1.11` | 4 of 4 replies through the inter-site link | 3 of 4 in the first attempt (first packet lost); 4 of 4 when repeated | Figure 7 | Passed |
| T-06 | PC-B1 | SRV-A (10.10.0.114) | `ping 10.10.0.114` | 4 of 4 replies | 4 of 4 replies | Figure 8 | Passed |
| T-07 | PC-B1 | SRV-A (10.10.0.114) | `tracert 10.10.0.114` | Path through Router-SedeB and Router-SedeA | 10.10.1.1 → 192.168.100.10 → 10.10.0.114 | Figure 8 | Passed |
| T-08 | PC-A1, PC-A2 | Router-SedeA (DHCP) | Set DHCP mode; `show ip dhcp binding` | Address from .11, gateway 10.10.0.1 | 10.10.0.11 and 10.10.0.12 | Figure 5 | Passed |
| T-09 | PC-B1, PC-B2 | Router-SedeB (DHCP) | Set DHCP mode; `show ip dhcp binding` | Address from .11, gateway 10.10.1.1 | 10.10.1.11 and 10.10.1.12 | Figure 5 | Passed |
| T-10 | SW-A1, SW-B1 | – | `show mac address-table` | MAC entries for the PCs and the router | Three entries in each switch | Figure 4 | Passed |
| T-11 | VLANs 10, 20, 30, 99 and 802.1Q trunk | – | – | Traffic separated by VLAN | Not implemented yet | _ | Pending |
| T-12 | Router-on-a-Stick | – | – | Routing between VLANs | Not implemented yet | _ | Pending |
| T-13 | EtherChannel and RSTP | – | – | Link aggregation and loop prevention | Not implemented yet | _ | Pending |
| T-14 | OSPF | – | – | Routes learned dynamically | Not implemented yet (temporary static routes in use) | _ | Pending |
| T-15 | NAT and firewall | – | – | Internet access and guest isolation | Not implemented yet | _ | Pending |


PC - A1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
C:\>ping 10.10.0.114
Pinging 10.10.0.114 with 32 bytes of data:
Reply from 10.10.0.114: bytes=32 time<1ms TTL=127
Reply from 10.10.0.114: bytes=32 time<1ms TTL=127
Reply from 10.10.0.114: bytes=32 time<1ms TTL=127
Reply from 10.10.0.114: bytes=32 time=21ms TTL=127
Ping statistics for 10.10.0.114:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 21ms, Average = 5ms
C:\>ping 10.10.1.11
Pinging 10.10.1.11 with 32 bytes of data:
Request timed out.
Reply from 10.10.1.11: bytes=32 time=8ms TTL=126
Reply from 10.10.1.11: bytes=32 time=1ms TTL=126
Reply from 10.10.1.11: bytes=32 time=1ms TTL=126
Ping statistics for 10.10.1.11:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 8ms, Average = 3ms
C:\>ping 10.10.1.11
Pinging 10.10.1.11 with 32 bytes of data:
Reply from 10.10.1.11: bytes=32 time=15ms TTL=126
Reply from 10.10.1.11: bytes=32 time=9ms TTL=126
Reply from 10.10.1.11: bytes=32 time=7ms TTL=126
Reply from 10.10.1.11: bytes=32 time=8ms TTL=126
Ping statistics for 10.10.1.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 7ms, Maximum = 15ms, Average = 9ms


PC - B1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.1.1
Pinging 10.10.1.1 with 32 bytes of data:
Reply from 10.10.1.1: bytes=32 time<1ms TTL=255
Reply from 10.10.1.1: bytes=32 time<1ms TTL=255
Reply from 10.10.1.1: bytes=32 time<1ms TTL=255
Reply from 10.10.1.1: bytes=32 time<1ms TTL=255
Ping statistics for 10.10.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.10.0.114
Pinging 10.10.0.114 with 32 bytes of data:
Reply from 10.10.0.114: bytes=32 time=8ms TTL=126
Reply from 10.10.0.114: bytes=32 time=1ms TTL=126
Reply from 10.10.0.114: bytes=32 time=2ms TTL=126
Reply from 10.10.0.114: bytes=32 time=4ms TTL=126
Ping statistics for 10.10.0.114:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 8ms, Average = 3ms
C:\>tracert 10.10.0.114
Tracing route to 10.10.0.114 over a maximum of 30 hops:
1 0 ms 0 ms 0 ms 10.10.1.1
2 0 ms 0 ms 0 ms 192.168.100.10
3 0 ms 0 ms 0 ms 10.10.0.114
Trace complete.


PC - A1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.0.1
Pinging 10.10.0.1 with 32 bytes of data:
Reply from 10.10.0.1: bytes=32 time=1ms TTL=255
Reply from 10.10.0.1: bytes=32 time<1ms TTL=255
Reply from 10.10.0.1: bytes=32 time<1ms TTL=255
Reply from 10.10.0.1: bytes=32 time<1ms TTL=255
Ping statistics for 10.10.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 10.10.0.12
Pinging 10.10.0.12 with 32 bytes of data:
Reply from 10.10.0.12: bytes=32 time<1ms TTL=128
Reply from 10.10.0.12: bytes=32 time<1ms TTL=128
Reply from 10.10.0.12: bytes=32 time<1ms TTL=128
Reply from 10.10.0.12: bytes=32 time<1ms TTL=128
Ping statistics for 10.10.0.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.10.0.114
Pinging 10.10.0.114 with 32 bytes of data:
Request timed out.
Reply from 10.10.0.114: bytes=32 time=8ms TTL=127
Reply from 10.10.0.114: bytes=32 time<1ms TTL=127
Reply from 10.10.0.114: bytes=32 time<1ms TTL=127
Ping statistics for 10.10.0.114:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 8ms, Average = 2ms
Activar Windows
Ve a Configuración para activar Windows.
Top


PC - A1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
C:\>ping 10.10.0.114
Pinging 10.10.0.114 with 32 bytes of data:
Request timed out.
Reply from 10.10.0.114: bytes=32 time=8ms TTL=127
Reply from 10.10.0.114: bytes=32 time<1ms TTL=127
Reply from 10.10.0.114: bytes=32 time<1ms TTL=127
Ping statistics for 10.10.0.114:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 8ms, Average = 2ms
C:\>ping 10.10.0.114
Pinging 10.10.0.114 with 32 bytes of data:
Reply from 10.10.0.114: bytes=32 time<1ms TTL=127
Reply from 10.10.0.114: bytes=32 time<1ms TTL=127
Reply from 10.10.0.114: bytes=32 time<1ms TTL=127
Reply from 10.10.0.114: bytes=32 time=21ms TTL=127
Ping statistics for 10.10.0.114:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 21ms, Average = 5ms
C:\>ping 10.10.1.11
Pinging 10.10.1.11 with 32 bytes of data:
Request timed out.
Reply from 10.10.1.11: bytes=32 time=8ms TTL=126
Reply from 10.10.1.11: bytes=32 time=1ms TTL=126
Reply from 10.10.1.11: bytes=32 time=1ms TTL=126
Ping statistics for 10.10.1.11:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 8ms, Average = 3ms
C:\>ping 10.10.1.11
Activar Windows
Ve a Configuración para activar Windows.
Top

## 8. Completion and Migration Plan

The prototype shows that addressing, DHCP and routing between sites work. VLANs with 802.1Q trunks, Router-on-a-Stick, EtherChannel, RSTP, OSPFv2, NAT, the firewall and the final OpenWrt configuration are still pending. They will be completed after the Week 10 content and the delivery of the equipment.

**What will replace the temporary parts**

| Temporary element in Phase 2 | Final element |
|---|---|
| **Serial link between routers** | Shared Wi-Fi network (192.168.100.0/24) through the wlan0 interface of each OpenWrt router |
| **Static routes** | OSPFv2 in Area 0 |
| **User and server networks connected to separate router ports** | VLANs 10, 20, 30 and 99 on an 802.1Q trunk, with Router-on-a-Stick |

**How the Cisco concepts translate to OpenWrt**

| Concept | Packet Tracer (Cisco) | OpenWrt |
|---|---|---|
| **Device name** | `hostname` | System configuration |
| **IP addressing** | `ip address` on each interface | Interfaces in the network configuration |
| **VLANs and trunk** | VLANs and trunk ports on the Cisco switches | VLAN interfaces on the router's wired port (for example eth0.10) |
| **Router-on-a-Stick** | Subinterfaces on one router port | One interface per VLAN, with the gateway address |
| **DHCP** | `ip dhcp pool` and excluded addresses | DHCP service (dnsmasq), with the range start used instead of excluded addresses |
| **Static routes** | `ip route` | Static routes in the network configuration |
| **OSPF** | Not yet used | FRRouting package (OSPFv2, Area 0) |
| **NAT** | Not yet used | Masquerading in the WAN firewall zone |
| **Firewall** | Not yet used | OpenWrt firewall zones and rules, deny by default |
| **EtherChannel and RSTP** | Not yet used | Not applicable: they stay on the Cisco switches |

**Remaining tasks and responsibilities**

| Task | Responsible | Support |
|---|---|---|
| VLANs 10, 20, 30, 99 and 802.1Q trunk on the switches | Samantha | Luis |
| Router-on-a-Stick | Luis | Samantha |
| EtherChannel and RSTP between the two switches of each site | Samantha | Luis |
| OSPF (Phase 3) | Luis | Samantha |
| NAT and firewall (guest isolation) | Luis | Samantha |
| OpenWrt installation on the Raspberry Pi | Luis | Jeniffer |
| Full tests, final test matrix and evidence | Jeniffer | Samantha |
| Report, repository and AI statement | Samantha | Jeniffer |

**Updated Schedule**

| Week | Activity |
|---|---|
| **9** | Packet Tracer prototype (this delivery) |
| **10** | Equipment delivery; VLAN and trunk content in class |
| **11** | Complete simulation (VLANs, Router-on-a-Stick, EtherChannel, RSTP, OSPF, NAT, firewall) and initial hardware deployment |
| **12–14** | Integration with the Go application and the FastAPI service; tests |
| **15** | Final review |
| **16** | Final presentation |

## 9. AI Use Statement and References

AI assistance (Claude, by Anthropic) was used as a guide to organize the work, suggest configuration commands and review draft text. The team built the topology in Packet Tracer, ran every command, took the screenshots and checked the results. The final text was written and reviewed by the team members.

## 10. References


Moy, J. (1998). *OSPF Version 2* (RFC 2328). Internet Engineering Task Force. https://www.rfc-editor.org/rfc/rfc2328

FRRouting. (n.d.). *FRRouting project*. https://frrouting.org/

OpenWrt Project. (n.d.). *OpenWrt documentation*. https://openwrt.org/docs/start

Universidad Internacional del Ecuador. (2026). *Proyecto Integrador – Tercer Nivel* [Project guide]. UIDE.